AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 31 · STUDENT LESSON

Design the Decision Process

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 10: AND, OR and ordered branches change outcomes.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Design a decision process that covers valid, invalid and no-match requests. Try the recall before opening notes.

Your learning target

I can design a decision **process** that covers valid, invalid and no-match requests.

I can trace changing **state** and explain why the search stops.

I can preserve both compatibility conditions and the agreed first-match rule.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

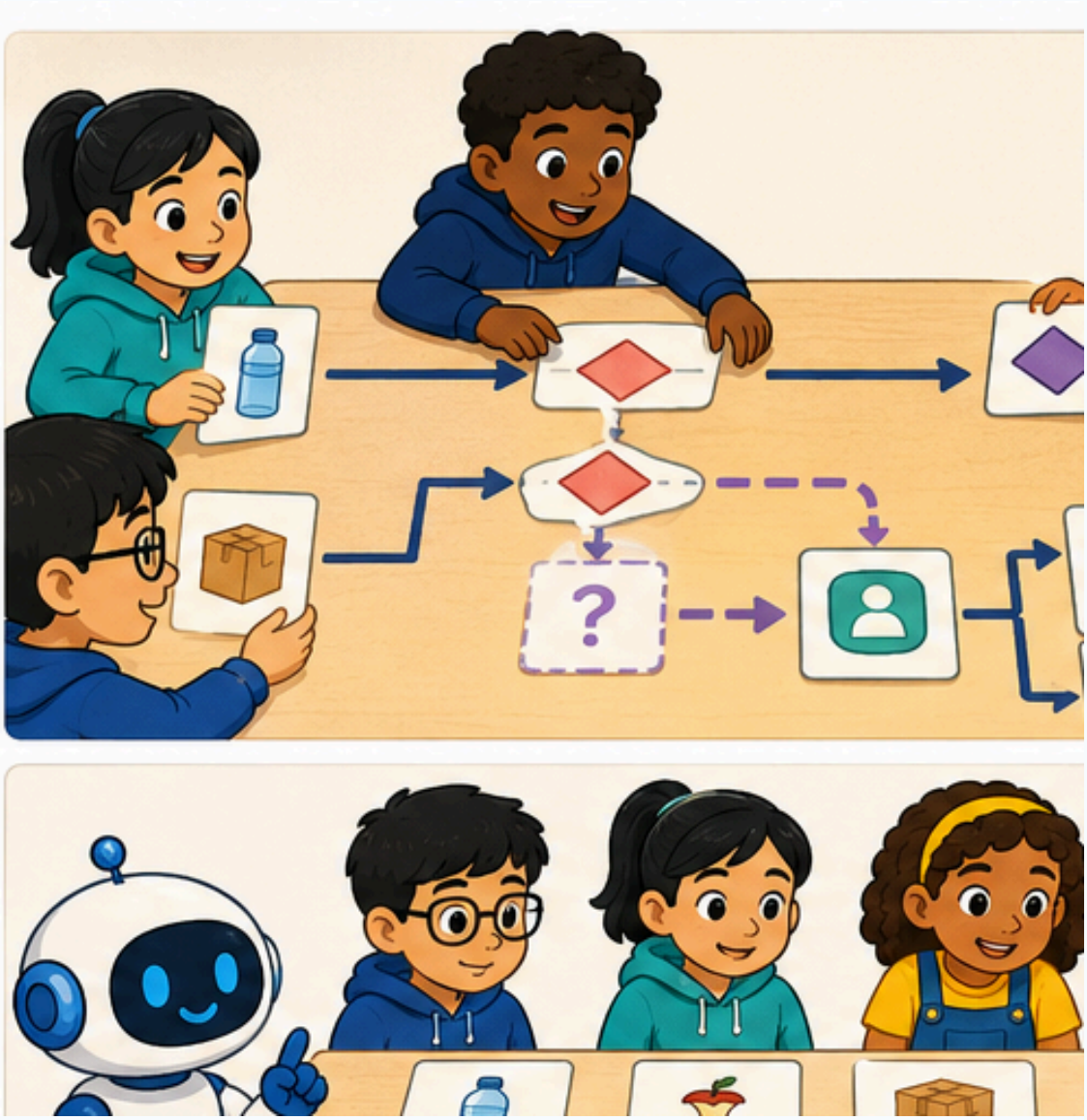
Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your **explanation** that the **evidence** shows was wrong.

Rules or learned patterns must connect **input**s to **output**s transparently.



Your target

LEARNING TARGET Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

Learner name:

Readiness for the connected explanation

For the activity project, how does a blank time differ from a valid request for zero minutes?

This week's end goal

By the end, I can design a complete **decision flow** with **validation**, search and stopping.

STUDENT LESSON · 2 OF 18

Notice the evidence

Investigate the supplied Design the Decision Process case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY A transparent sorter first checks item type, then one disambiguating feature, then sends unclear items to review.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and **state** the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A complete **process** says what happens for every kind of request in the task. It must handle a **fitting** activity, no **fitting** activity and unusable **input**. We will design a paper algorithm before building the prototype.

Its decisions use the requirements from Week 30, so a different drawing should not silently change what counts as correct.

Keep the same catalogue order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These are fixed paper-model values with materials assumed available. Valid **input** is a whole number of minutes from 0 to 30 and Quiet required Yes or No.

A **candidate** fits when its required minutes are no more than the available minutes and either quiet is not required or the activity is quiet. Both parts must hold.

The **output** is the first **fitting** activity in catalogue order, with a reason, or a clear no-match result. Invalid or missing **input** produces a clarification request. None of these **output**s books materials or publishes a message.

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Design the Decision Process. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.



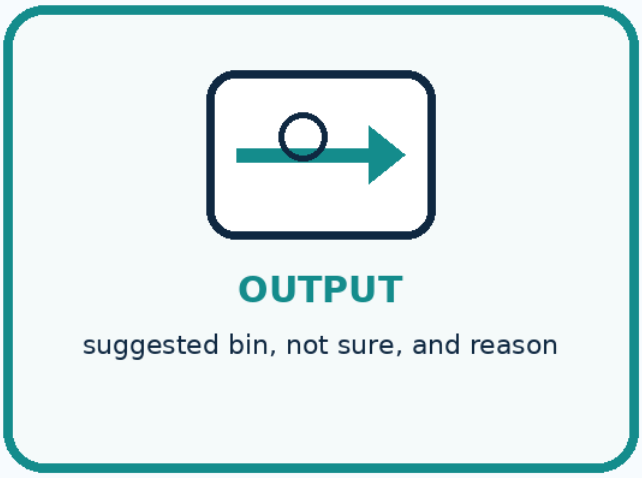
INPUT

approved input fields



PROCESS

smallest rule or learned decision flow with visible branches



INPUT approved **input** fields

PROCESS smallest rule or learned **decision flow** with visible **branch**es

OUTPUT suggested bin, not sure, and reason

Key idea

CORE IDEA Rules or learned patterns must connect **input**s to **output**s transparently.

How the mechanism works

UNDER THE HOOD A decision **process** operationalizes the interface contract. Simplicity improves testability and **explanation**, but oversimplification can erase relevant distinctions.

Learned components need explicit data, threshold, and fallback documentation.

Draw a fresh example with all three stages:

A closer explanation

Four paths through the recommender

CONDITION REACHED	NEXT ACTION	WHERE THE PATH ENDS
Input missing or invalid	Ask for corrected input	Stop this run
Candidate fails a check	Move to the next row	Continue search
Candidate passes both	Suggest it with a reason	Stop this run
All rows checked and failed	Report no match	Stop this run

Failing one candidate continues the search. No match is justified only after the valid request has no fitting candidate.

Begin by checking the request. If minutes are missing, outside 0 to 30 or not a whole number, ask for a whole number in that range. If the quiet field is not Yes or No, ask for one of those values.

State the fields needing correction and stop this run. When a corrected request arrives, validate it as a new run rather than continuing with old **candidate state**.

For a valid request, start at B1. Check time and the quiet condition for the current **candidate**. If both pass, **output** that **candidate** and stop. If either fails, move forward exactly one row.

After B4 fails, there is no next row, so **output** no match and stop. Never jump back to B1 within the same search.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

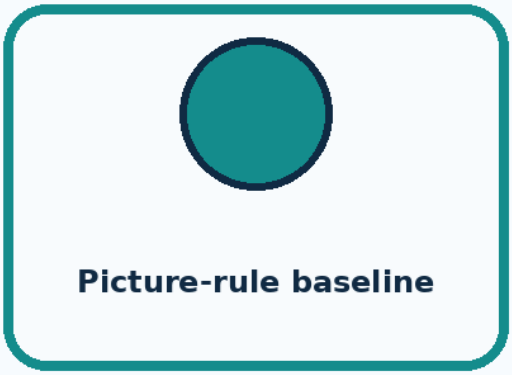
Compare the cases

> Compare the supplied cases for Design the Decision Process. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — Picture-rule **baseline** Clue: direct visible **branch**es. Meaning: easy to inspect.

The case to investigate

CASE 2 — Learned classifier Clue: patterns from examples. Meaning: needs data **evidence**.

The case to investigate

CASE 3 — Extra unused field Clue: never changes path. Meaning: remove.

The case to investigate

CASE 4 — No fallback Clue: forces ambiguous label. Meaning: add review.

Keep this boundary clear

BOUNDARY Transparent does not mean correct; the flow still needs testing.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: Increase scissors needed to 3.

Predicted result

Prediction: K3 is selected.

Why the result follows

Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Design the Decision Process to explain a concrete decision, rather than only memorising definitions.

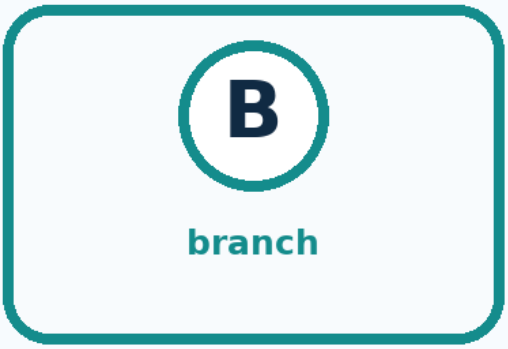
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
decision flow	ordered path connecting inputs to outputs	
baseline	simple method used for comparison	

Word	Meaning	My example
branch	path chosen by a condition	
explanation	traceable reason for an output	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR A longer flow must be more capable.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Validation	Checking an input against the agreed rules before using it
Candidate	An option currently being checked
State	Information that records the process's current position or values
Termination	Reaching a point where the process stops

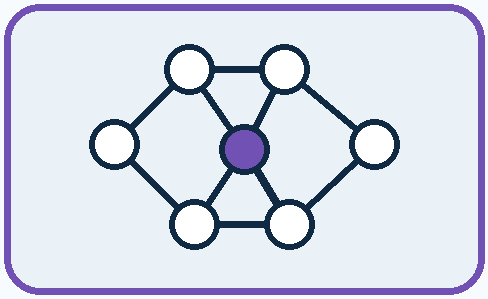
Follow a worked example

> Follow the worked trace for Design the Decision Process, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE A transparent sorter first checks item type, then one disambiguating feature, then sends unclear items to review.

Step 1

STEP 1 — OBSERVE approved **input** fields

Step 2

STEP 2 — TRACE smallest rule or learned **decision flow** with visible **branch**es

Step 3

STEP 3 — CONCLUDE suggested bin, not sure, and reason

Model answer

MODEL REASONING Start with a **baseline** whose **branch**es can be read aloud. Each **input** must affect a decision or be removed. Add not-sure and review paths, then explain when a learned component would earn its added complexity.

A claim to avoid

DO NOT OVERCLAIM Transparent does not mean correct; the flow still needs testing.

What new **evidence** could change this answer?

A closer explanation

Stage for 9 minutes and Quiet required No	Evidence	Next state or output
Validate	9 is a whole number in range; No is valid	Current candidate B1
Check B1	10 minutes is more than 9	Move to B2
Check B2	8 is no more than 9; quiet is not required	Suggest B2 and stop

Worked explanation · Design the Decision Process

The current **candidate** changes from B1 to B2. The request values stay the same. After B2 passes, the **process** stops: it does not replace B2 with a later **fitting** row.

This early stop preserves the first-match rule and gives the same required suggestion as the manual **baseline**, which marked every row before selecting the first fit.

Worked explanation · Design the Decision Process

Termination needs a reason. There are only four rows, each failed check moves to the next row, and no row is revisited. A valid request therefore examines at most four **candidate**s. A success can stop sooner.

Invalid **input** stops before the search. Merely drawing an arrow around a loop is not enough; the **state** must make progress toward an end.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Design the Decision Process

A fictional worksheet routes craft kits by required glue and number of scissors.

The facts we will use

Supplied facts

K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4.

Supplied facts

Request: glue required Yes, scissors needed 2. Return first kit meeting both requirements.

AI ACADEMY / AI-1/31

Design the Decision Process

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate fields

Require Yes/No glue input and whole scissors count 0-4.



2. Evaluate K1

Scissors fit, but required glue is absent: continue.



3. Evaluate K2

Glue is present and 2 scissors meet the request: accept.



4. Stop

Return K2 without replacing it with later K3.

CHANGE ONE THING

Increase scissors needed to 3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate fields	Require Yes/No glue input and whole scissors count 0–4.
2. Evaluate K1	Scissors fit, but required glue is absent: continue.
3. Evaluate K2	Glue is present and 2 scissors meet the request: accept.
4. Stop	Return K2 without replacing it with later K3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

What this example does not establish

Keep the stated limit

A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

Keep the stated limit

Read the labelled picture: Where does the first original rejection occur, and why is K3 not originally selected?

Keep the stated limit

Change one thing: Increase scissors needed to 3. Predict the result before checking the **explanation**. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Design the Decision Process claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Picture-rule baseline	direct visible branches	easy to inspect
Learned classifier	patterns from examples	needs data evidence
Extra unused field	never changes path	remove
No fallback	forces ambiguous label	add review

Repair two tempting claims

Challenge claim A

CLAIM A A longer flow must be more capable.

Repair:

Challenge claim B

CLAIM B A **learned model** does not need a visible fallback or **explanation**.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

An **explanation** should identify the checks that actually justified the result. For example, B2 fits the 9-minute No request because its 8-minute duration fits and quiet was not required.

Do not add “you will love it”, which was neither an **input** nor a tested condition. For no match, report that no catalogue row met both conditions; a trace can show why.

Check the claim against the facts

Save a labelled version of the **process** with its **validation**, **candidate** order, conditions, **state** changes and stopping points. A partner should be able to follow it without asking you to fill a missing step.

Next week the paper prototype will turn this design into instruction cards and a trace record. Keep any final evaluation cases reserved while developing the design.

A design finds B1 compatible for 15 minutes No but keeps searching and returns B4 instead. Is B4 compatible? Does this output meet the full task requirements? Explain the difference and give the required output

STUDENT LESSON · 8 OF 18

Practise together

Practise Design the Decision Process with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Rules or learned patterns must connect **input**s to **output**s transparently.

Use a fresh example to explain **decision flow**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A **learned model** does not need a visible fallback or **explanation**.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Consider the mistaken instruction “If B1 fails, report no match.” It would miss B2 on the 9-minute No request.

Another mistake is “If time fits or quiet fits, suggest the activity.” That could allow an activity that takes too long merely because it is quiet. Use a concrete request to show which requirement a design breaks.

> **Trace 9 minutes with Quiet required Yes through every candidate. State the reason each fails and the final output. Then explain what would happen if a failed candidate were checked again without moving to the next row. Finally, state the first step for 31 minutes with Quiet required No**

A **process** must also distinguish the meaning of No from a requirement for noise. When Quiet required is No, a quiet **candidate** can still pass. For 11 minutes No, B1 fits and comes first, even though B2 also fits.

An instruction to reject all quiet activities when No is selected would change the task.

Practise with less help: Track the current candidate

Use workbook W2 for the connected example "Designing a complete decision **process**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own **explanation**.

Check your reasoning

Use the worked case for Designing a complete decision **process**. Proposed **explanation**: "One failed row proves no match". Decide whether this **explanation** is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Where does the first original rejection occur, and why is K3 not originally selected?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

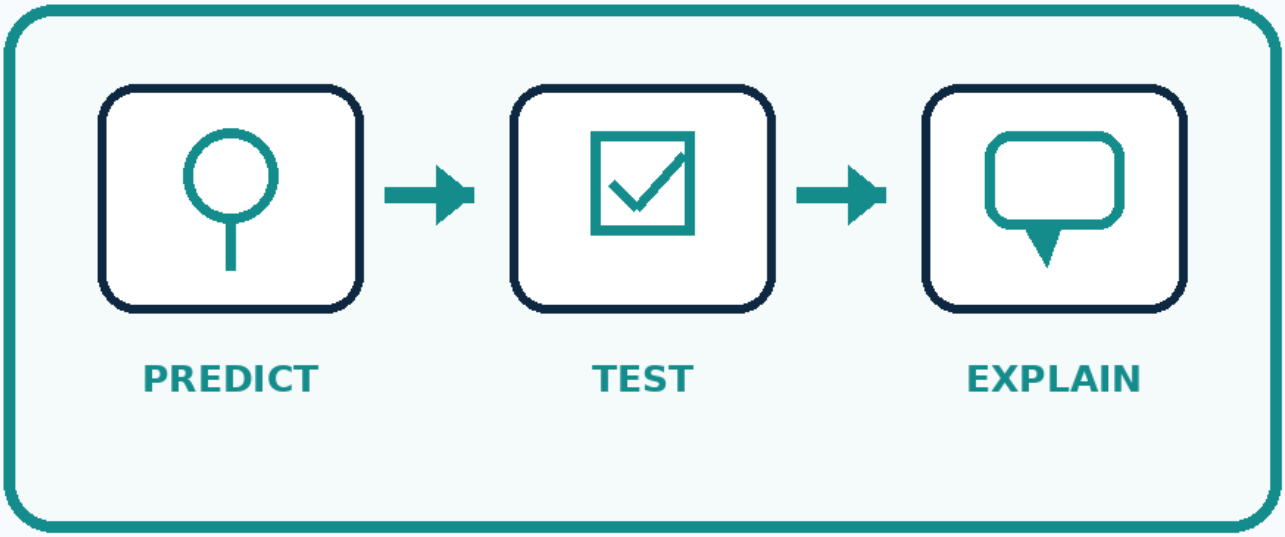
Run the investigation

> Investigate Design the Decision Process. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Smallest Useful Flow

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Create and trace the smallest useful **decision flow**.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

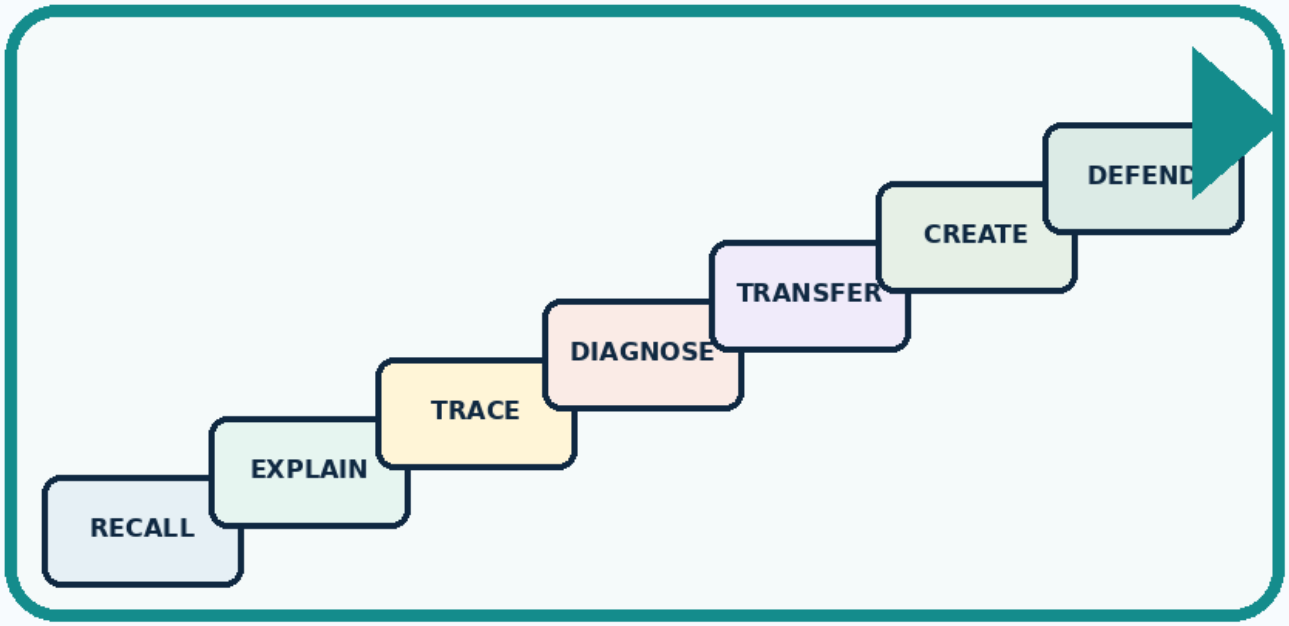
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Design the Decision Process. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **decision flow**.

EXPLAIN Rules or learned patterns must connect **input**s to **output**s transparently.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: A longer flow must be more capable.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

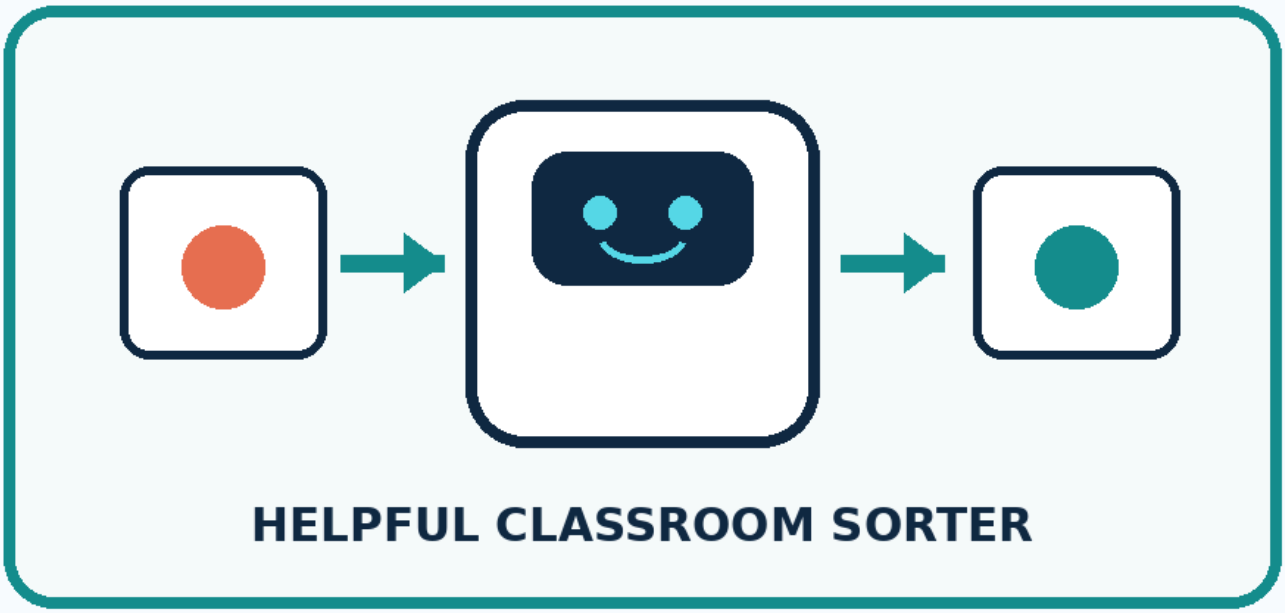
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Complete the sorter v1 **decision flow** and annotate every **branch** with the **evidence** it uses.

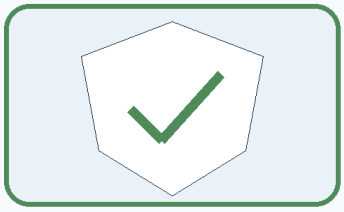
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Transparent does not mean correct; the flow still needs testing.

My project decision:

Evidence for your project

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Create and trace the smallest useful **decision flow** across normal, missing, and ambiguous **input**s.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ **State** the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Start with a **baseline** whose **branch**es can be read aloud. Each **input** must affect a decision or be removed. Add not-sure and review paths, then explain when a learned component would earn its added complexity.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Build a Paper Prototype.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Design the Decision Process case for supported practice. Repeated examples do not become fresh assessments.

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

The question

Trace glass and repair the rule.

Worked explanation

Glass incorrectly goes to M. Add an explicit metal check and a glass or review **branch**. Otherwise includes all remaining cases.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Otherwise includes every remaining case

Evidence or stage	Value or result
Paper	Yes → P
Not paper, metal	Yes → M
Not paper, not metal	Glass or review

A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

Trace glass and repair the rule.

Glass incorrectly goes to M. Add an explicit metal check and a glass or review **branch**. Otherwise includes all remaining cases.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Design a complete room decision process

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 31 Independent application

Close the worked **explanation**. Apply the idea to this different question without copying.

Write a decision **process**: a clear pencil goes to the pencil tray, a clear book to the shelf, and an unknown object to review. Trace a lunch box.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Complete the sorter v1 **decision flow** and annotate every **branch** with the **evidence** it uses.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Design the Decision Process, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional worksheet routes craft kits by required glue and number of scissors.

By the end, I can	How I will show it
Design a decision process that covers valid, invalid and no-match requests	Explain the deciding step, show a trace or test, and state one limit.
Trace changing state and explain why the search stops	Explain the deciding step, show a trace or test, and state one limit.
Preserve both compatibility conditions and the agreed first-match rule	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Design the Decision Process without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Design a decision **process** that covers valid, invalid and no-match requests

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional worksheet routes craft kits by required glue and number of scissors.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Return K2 without replacing it with later K3.

1. Stop

2. Evaluate K1

3. Evaluate K2

4. Validate fields

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: Stop Return K2 without replacing it with later K3. A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase scissors needed to 3. Which conclusion is supported by the supplied facts?

1. K3 is selected.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported **explanation**: K3 is selected. K1 lacks glue;

> K2 has only 2 scissors; K3 has glue and 4.

Repair a missing building block

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same **explanation**?.

Return to this week and explain what you changed.

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two need separate approval?. Return to this week and explain what you changed.

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

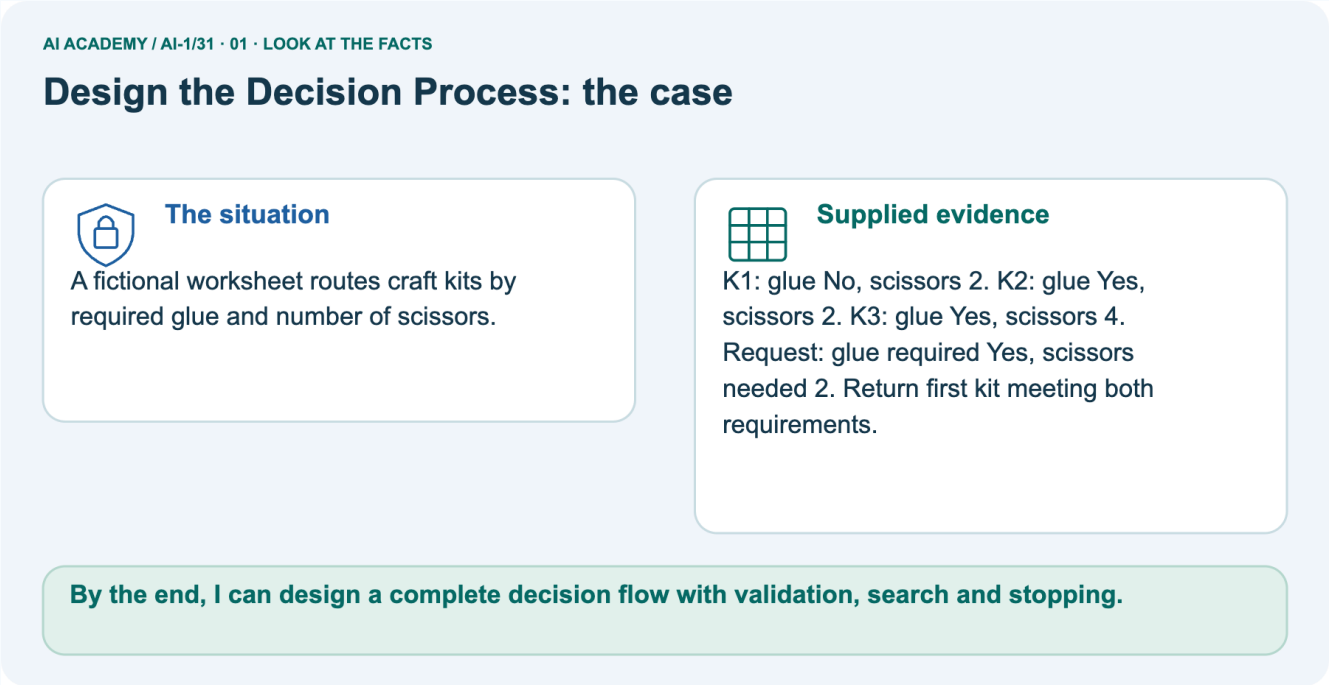
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Design the Decision Process graphic using its labels, deciding evidence and relevant limitation.

01. Design the Decision Process: the case



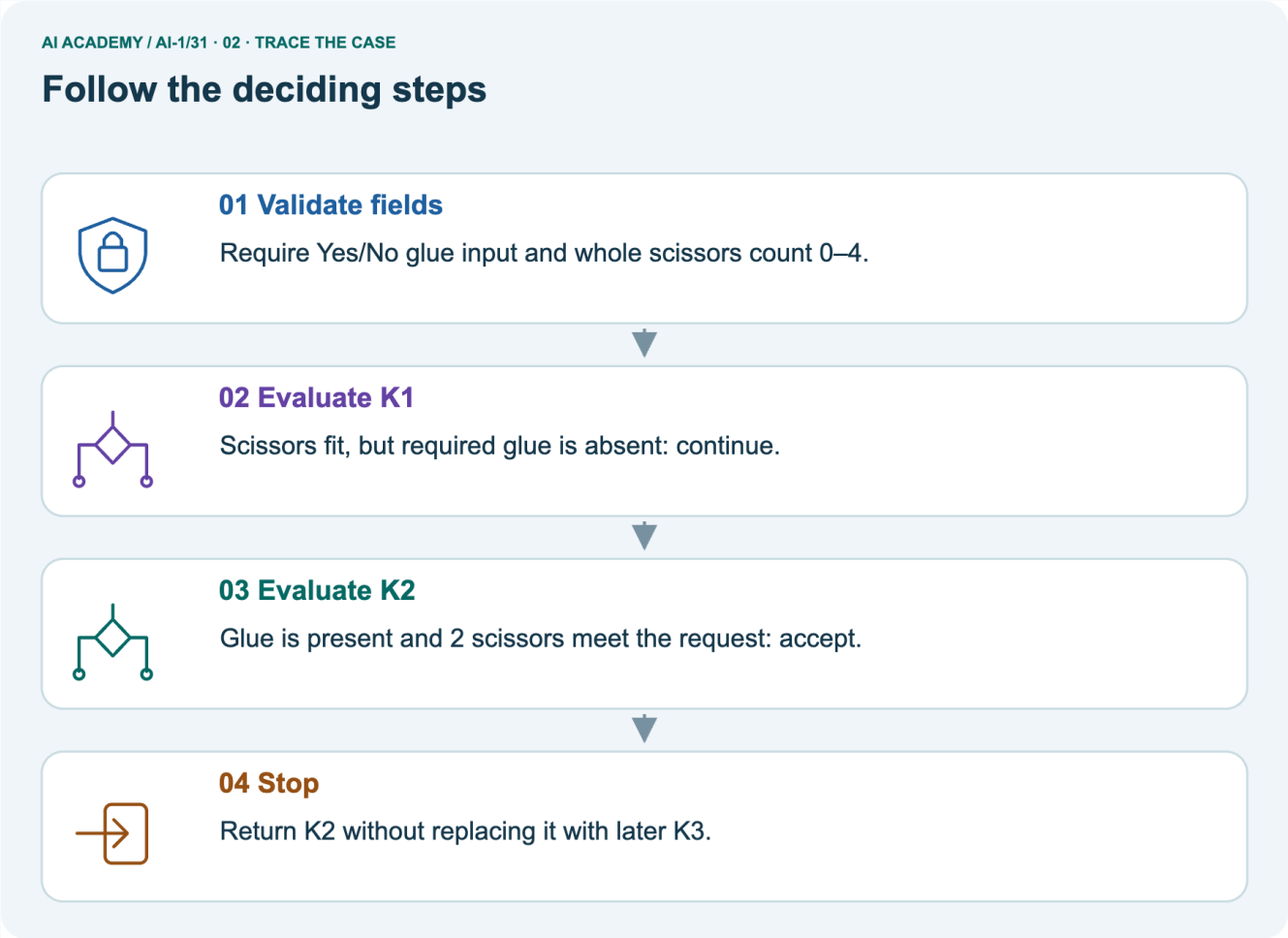
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional worksheet routes craft kits by required glue and number of scissors. Supplied facts: K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4. Request: glue required Yes, scissors needed 2.

Return first kit meeting both requirements.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate fields: Require Yes/No glue **input** and whole scissors count 0–4. → Evaluate K1: Scissors fit, but required glue is absent: continue. → Evaluate K2: Glue is present and 2 scissors meet the request: accept. → Stop: Return K2 without replacing it with later K3.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Four paths through the recommender

AI ACADEMY / AI-1/31 · 03 · INSPECT THE LESSON MODEL

Four paths through the recommender

Condition reached	Next action	Where the path ends
Input missing or invalid	Ask for corrected input	Stop this run
Candidate fails a check	Move to the next row	Continue search
Candidate passes both	Suggest it with a reason	Stop this run
All rows checked and failed	Report no match	Stop this run

Failing one candidate continues the search. No match is justified only after the valid request has no fitting candidate.

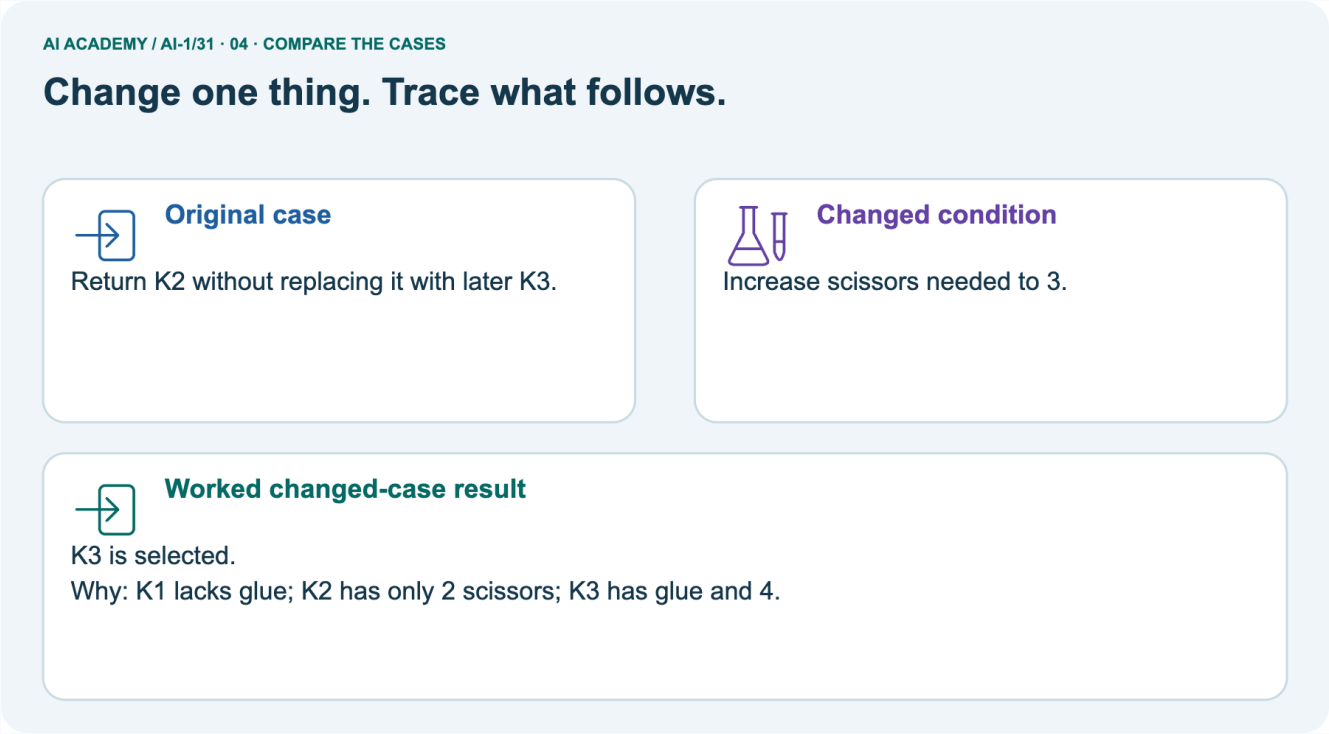
Follow the labelled picture with the matching explanation. Open the vector diagram

Four paths through the recommender. **Input** missing or invalid | Ask for corrected **input** | Stop this run. **Candidate** fails a check | Move to the next row | Continue search.

Candidate passes both | Suggest it with a reason | Stop this run. All rows checked and failed | Report no match | Stop this run

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

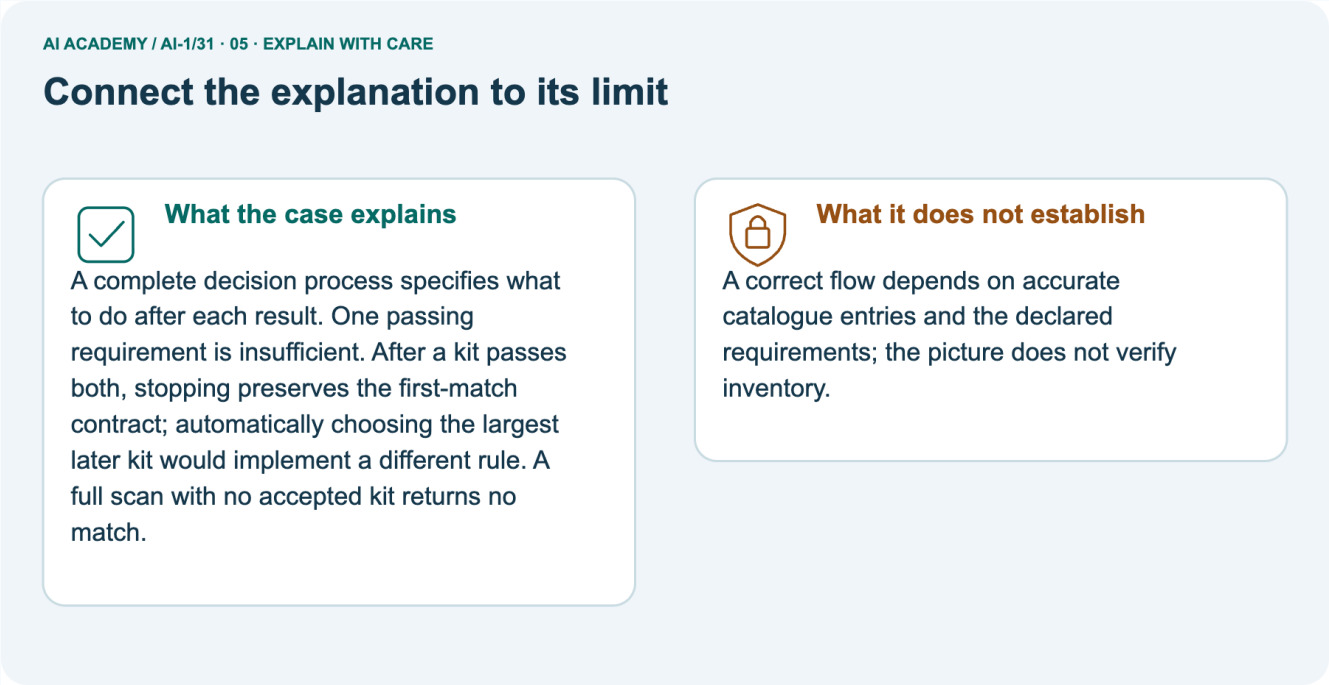


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return K2 without replacing it with later K3. Changed condition: Increase scissors needed to 3. Expected result: K3 is selected. Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

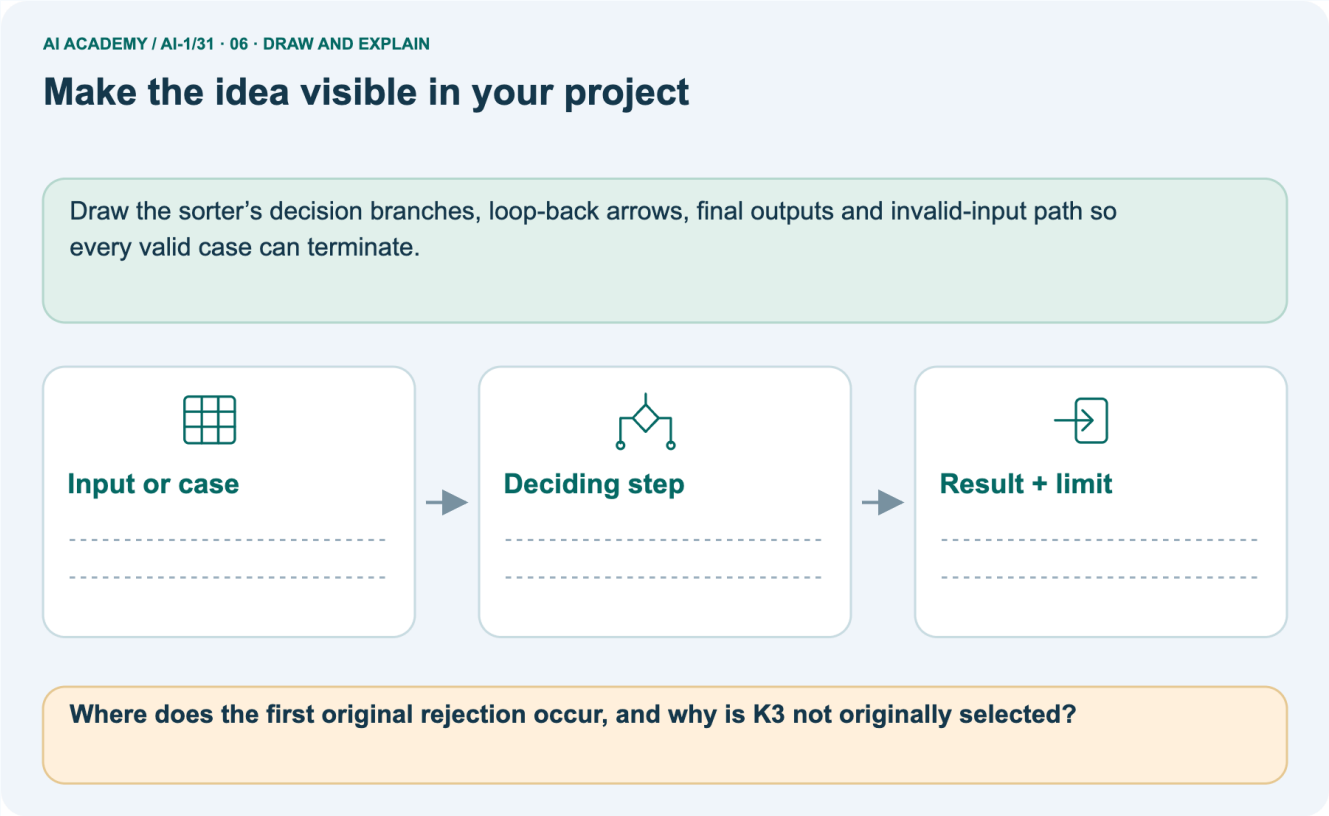
A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Limit: A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate. Draw your **input**, deciding step, result and limit.

Supported question: Where does the first original rejection occur, and why is K3 not originally selected?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.